

What Makes a Fortnightly Lookback Work? Spectral Triangulation of Tidal Structure in Daily Estuarine Salinity

Khang D. Nguyen^{1,2}, Luong Thanh Tra Ngo³^(✉), and
Khuong Nguyen-An^{1,2}^(✉)

¹ Ho Chi Minh City University of Technology (HCMUT), 268 Ly Thuong Kiet
Street, Dien Hong Ward, Ho Chi Minh City, Vietnam

² Vietnam National University Ho Chi Minh City, Linh Xuan Ward, Ho Chi Minh
City, Vietnam

Emails: {khang.nguyen4growth, nakhuong}@hcmut.edu.vn

³ FPT University, Greenwich Vietnam, Ho Chi Minh Campus, Vietnam
Email: TraNLT2@fpt.edu.vn

Abstract. Machine-learning forecasts of estuarine salinity require a lookback window, and current practice fixes its length by search. Recent work on the Vietnamese Mekong Delta converges on roughly a fortnight, attributing the choice to an unspecified periodicity; whether that number is physical or merely an empirical optimum determines whether it transfers across sites. We apply a four-stage spectral triangulation protocol to a sixteen-year record of daily maximum salinity from the Tien River estuary. The deseasonalised spectrum contains not one periodicity but a resolved triplet: a dominant peak at 14.77 d exceeding a family-wise permutation threshold by a factor of 2.5 under two decomposition models, flanked by significant peaks at 14.19 d and 15.39 d. These are the only peaks in the 5 d to 100 d band clearing even the 95th percentile, and their positions are fixed to within one spectral bin by astronomical constants alone. Two routes predict exactly those positions — annual amplitude modulation of the fortnightly tide, and aliasing of the diurnal, semi-diurnal and shallow-water constituents under daily sampling — and we show they are the same arithmetic statement, hence not separable at daily resolution; phase-randomised surrogates cannot adjudicate either.

Keywords: Time series analysis · Spectral estimation · Seasonality diagnostics · Tidal aliasing · Amplitude modulation · Estuarine salinity.

1 Introduction

Data-driven salinity forecasting in the Vietnamese Mekong Delta has settled into a recognisable pattern: assemble a multi-year record, select a family of learners, and fix the input window by searching a small grid. Dang et al. [2] read a cyclical repetition at seven to eight steps off the correlograms of two-hourly salinity and adopted the corresponding 16 h horizon, reporting that differences

across candidate horizons were not statistically significant, and justifying the choice on operational rather than evidential grounds. A later study at daily resolution searches lookback lengths of 15, 30, 45, and 60 days across two Cung Hau estuary stations and reports the shortest, 15 days, as best overall [12]; an explanation is gestured at — shorter windows are said to capture “the periodicity of salinity intrusion” — but that periodicity is never identified, no frequency-domain analysis is performed, and the authors state that no statistical testing was applied. Both studies reach the same position from different data: a window length is selected, a periodic mechanism is invoked in general terms, and the connection is left unmade.

The gap matters because the two kinds of numbers behave differently. One obtained by grid search is a property of one dataset, station, and model family; one corresponding to a known physical periodicity transfers to sites under the same forcing and can be stated in advance. The question is therefore narrow and answerable: *does the fortnightly window that empirical search keeps selecting correspond to an identifiable periodicity in the data, and if so, what is the actual spectral structure at that scale?*

The obvious route fails: on the raw daily series the annual cycle and day-to-day persistence bury any fortnightly structure (Section 6.1) — which is what produced the ambiguous reading of [2]. We therefore triangulate: unit-root testing establishes which treatment is licensed, decomposition removes the annual component, spectral estimation with an explicit null identifies what survives multiplicity control, and a parametric prediction from astronomical constants, fixed before the spectrum is consulted, provides an external check.

We make the following contributions.

- The deseasonalised spectrum contains a resolved triplet at 14.77 d, 14.19 d and 15.39 d; the dominant member exceeds a family-wise permutation threshold by $2.54\times$ and $2.52\times$ under multiplicative and additive decomposition; and these three are the only peaks in the 5 d to 100 d band clearing even the 95th percentile (Section 6.2).
- All three positions are fixed, to within one spectral bin, by astronomical constants with no fitted parameters (Sections 4.3 and 6.3).
- The two candidate explanations for those positions — annual amplitude modulation of the fortnightly tide, and aliasing of O_1 , M_2 and MK_3 under daily sampling — are not competing hypotheses but the same arithmetic statement, because the identities $M_2 = O_1 + K_1$ and $MK_3 = M_2 + K_1$ survive aliasing while K_1 aliases onto the annual frequency. The mechanism is consequently not decidable at daily resolution, whatever the quality of the fit (Sections 4.2 and 6.3).
- The standard diagnostic battery applied to the raw series — autocorrelation, partial autocorrelation, and the unwindowed periodogram — recovers none of this structure, for three distinct reasons (Section 6.1).
- Phase-randomised surrogate testing is structurally incapable of validating amplitude modulation, because a spectrum-preserving null necessarily

contains the effect under test; we report our own non-significant outcome as an illustration (Section 7.1).

2 Related Work

We first fix notation. For a univariate series $\{x_t\}$ at a fixed interval — here daily salinity — a forecasting model is a map $\hat{x}_{L+1:L+H} = f_\theta(x_{1:L})$ with $x_{1:L} = \{x_t\}_{t=1}^L$, predicting H future values from the L most recent observations. The integer L is the *lookback window*, a property of the problem specification rather than of the learner: before any θ is estimated it fixes which temporal scales are representable at all, since a component of period T_0 is recoverable from within the window only if $L \gtrsim T_0$ and slower components are inaccessible in principle. This paper is concerned exclusively with L ; we do not estimate f_θ , compare model families, or report forecast error.

Recent applied work in the delta has concentrated on model comparison. Duc et al. [4] evaluate six learners and find architectures with explicit sequence memory most accurate, with partial dependence reproducing the expected spatial gradient; Dang et al. [2], on short horizons at coastal stations, report the opposite ordering. Nguyen et al. [12] instead isolate temporal context itself, crossing lookback lengths of $\{15, 30, 45, 60\}$ days with one- to three-day horizons for a univariate LSTM, and select 15 days without locating the periodicity their explanation invokes. More broadly, frequency-domain structure routinely constrains a model’s input representation — through basis expansions on intrinsic frequencies [13], Fourier-identified dominant periods [23], frequency-basis sparsity [24], or past intervals selected from the dominant cycles of an extracted seasonal component [7] — but always locating periodicities empirically and treating what is found as the operative structure. *What remains open*: whether the window lengths repeatedly selected by search correspond to an identifiable periodicity of the estuarine system, and whether such a period can be cross-validated against an independent, non-data-derived prediction of where it should lie.

Two further bodies of literature bear on that question. In econometrics, the augmented Dickey–Fuller [3] and KPSS [10] tests address a unit root at the zero frequency and are silent about seasonal frequencies, while seasonal unit-root tests — OCSB [14] and Canova–Hansen [1] — address whether the seasonal pattern is itself stochastic, with the sequential protocol between them set out in [9]; applied hydrological studies rarely report the seasonal diagnostics, so the licence for differencing versus decomposition is usually left implicit. In tidal science, classical harmonic analysis decomposes the water level into constituents at astronomically determined frequencies [16], the relevant result being that the spring-neap cycle is a beat between the principal lunar and solar semi-diurnal constituents [18]; since intrusion length is governed by tidal amplitude [11] salinity should inherit that rhythm, and tidal amplification is documented at Tien River mouth stations [5]. Conducted at sub-daily resolution, this work does not address what survives daily sampling — which is itself well understood next

door, since satellite altimetry samples at repeat periods of days to weeks, every constituent is aliased, and the orbit design is partly a matter of placing those aliases harmlessly [15], the resulting alias periods and identities among them being routine [19]. *What remains open*: those relations, standard in altimetry and applied in Section 4.2, appear not to have reached the literature in which daily-sampled estuarine records are used to choose forecasting input windows.

3 Data and Study Site

The record is a daily series from the Vam Kenh station on the Tien River branch of the Mekong Delta, Vietnam, provided by the Tien Giang Irrigation Works Operating One Member Company Limited. Vam Kenh sits at the seaward end of the branch, among the stations documented as experiencing progressive increases in salt intrusion [5], so tidal forcing there should be strong relative to interior stations. Each daily value is the *maximum* salinity of that calendar day: the quantity of operational interest, since peak intrusion rather than the daily average determines whether an intake can be operated, and — being an order statistic rather than a point sample — a nonlinear functional of the tide, which Section 4.2 takes up. The series spans 2009-01-01 to 2024-12-31, giving $N = 5,844$ observations with no missing values; mean 12.36 g L^{-1} (s.d. 6.12), interquartile range 6.90–17.80, minimum 1.90, maximum 27.00 g L^{-1} , with a longer upper than lower tail, consistent with the physical asymmetry of the process.

Completeness results from gap-filling via monotonicity-preserving cubic interpolation [6], but the analysed window required very little of it. The full station record spans 2000–2024 and is 11.49 % incomplete, yet this is concentrated in the early years: over 2000–2008, excluded here, 28.98 % of days are missing, whereas the analysed window 2009–2024 is 98.36 % complete, with only 96 missing observations in 32 gaps, none longer than eleven days.

4 Theory: Beat, Aliasing and Modulation Sidebands

This section derives, from astronomical constants alone, the periods the physics predicts should appear in a daily record — by two independent routes that turn out to give the same three numbers.

4.1 The fortnightly beat

In the mixed semi-diurnal regime of the East Sea the dominant constituents are the principal lunar semi-diurnal M_2 ($T_{M_2} = 12.4206 \text{ h}$) and the principal solar semi-diurnal S_2 ($T_{S_2} = 12.0000 \text{ h}$) [16,18]. The spring-neap cycle is their beat: superposed at equal amplitude they give an envelope $\cos[(\omega_{M_2} - \omega_{S_2})t/2]$ on a carrier, and the interval between envelope maxima defines

$$T_{M_{sf}} = \left(\frac{1}{T_{S_2}} - \frac{1}{T_{M_2}} \right)^{-1} = \left(\frac{1}{12.0000} - \frac{1}{12.4206} \right)^{-1} \text{ h} = 14.765 \text{ d}, \quad (1)$$

Table 1. Alias periods under once-daily observation, from the standard constituent speeds: $1/|a(f)|$ is the period over which the phase drifts through a full cycle relative to the calendar day.

Constituent Description		Period (h)	sgn $a(f)$	$ a(f) $ (d^{-1})	Alias (d)
S_2	principal solar semi-diurnal	12.0000	—	0.000000	— (d.c.)
M_2	principal lunar semi-diurnal	12.4206	—	0.067726	14.7653
K_1	lunisolar declinational diurnal	23.9345	+	0.002738	365.243
O_1	principal lunar diurnal	25.8193	—	0.070464	14.1916
MK_3	shallow-water compound $M_2 + K_1$	8.1771	—	0.064988	15.3873

the lunisolar synodic fortnightly constituent M_{sf} . Since intrusion length scales with tidal amplitude [18,11] and the envelope modulates precisely that amplitude, salinity at a seaward station should inherit a periodicity at $T_{M_{sf}}$.

4.2 Aliasing under daily observation

A daily interval places the Nyquist frequency at 0.5 d^{-1} , so no constituent is resolved. What survives is its phase relative to the calendar day: a constituent of frequency f (cycles per day) advances in phase by $2\pi f$ daily, hence by $2\pi a(f)$ modulo 2π , where

$$a(f) = f - \text{round}(f). \quad (2)$$

Any once-daily observable depending on that phase — a point sample, a daily mean, or, as here, a daily maximum — is modulated at period $1/|a(f)|$, which we call the alias period, noting that for a nonlinear daily statistic it arises from phase drift rather than aliasing in the textbook sense. Table 1 evaluates it.

Three features drive the rest of the paper. First, S_2 aliases exactly onto zero frequency and is indistinguishable from a constant. Second, the alias of M_2 falls at 14.7653 d — *exactly* the spring-neap period of (1), since with S_2 at zero that alias is by construction the difference frequency $f_{M_2} - f_{S_2}$. Third, and less obviously, aliasing is linear modulo unity, so alias frequencies inherit the additive identities among constituent frequencies: from $M_2 = O_1 + K_1$ and $MK_3 = M_2 + K_1$,

$$a(O_1) = a(M_2) - a(K_1), \quad a(MK_3) = a(M_2) + a(K_1). \quad (3)$$

Both hold for the *signed* quantity of (2); since $a(M_2)$ and $a(K_1)$ carry opposite signs, in the tabulated magnitudes they read $|a(O_1)| = |a(M_2)| + |a(K_1)|$ and $|a(MK_3)| = |a(M_2)| - |a(K_1)|$, the operations exchanging places when the sign is dropped. And K_1 , whose frequency is that of the sidereal day, aliases to 365.243 d — the year, because the difference between sidereal and solar rotation rates accumulates to one cycle annually. In the aliased domain O_1 , M_2 and MK_3 therefore sit at $f_{M_{sf}} + f_{\text{ann}}$, $f_{M_{sf}}$ and $f_{M_{sf}} - f_{\text{ann}}$: an evenly spaced triplet about the fortnightly period, separated by the annual frequency, produced by aliasing alone and requiring no modulation whatever. One remark is specific to

the daily maximum: being an order statistic it rectifies the tide, and rectification generates difference frequencies directly, the leading one being $f_{M_2} - f_{S_2}$ again. The nonlinearity therefore reinforces the fortnightly component at exactly the period the phase-drift argument already predicts.

4.3 Modulation sidebands

Proposition 1. *If a carrier of frequency f_c has its amplitude modulated by a sinusoid of frequency f_m with depth m , so that $x(t) = [1 + m \cos(2\pi f_m t)] \cos(2\pi f_c t)$, then the spectrum of x consists of exactly three components, at f_c , $f_c - f_m$ and $f_c + f_m$, with amplitude ratio $1 : m/2 : m/2$.*

Proof. Expand and apply $\cos \alpha \cos \beta = \frac{1}{2}[\cos(\alpha - \beta) + \cos(\alpha + \beta)]$ to the product term; the three sinusoids have the stated frequencies and amplitudes. $\square$

If the fortnightly influence on salinity is modulated over the annual cycle — as the hydrology suggests, since discharge governs how far tidal forcing penetrates — the spectrum should contain three peaks, symmetric in frequency about the carrier. With $f_c = 1/T_{M_{sf}}$ from (1) and $f_m = 1/365.25$ d (tropical year),

$$T^+ = \left(\frac{1}{14.765} + \frac{1}{365.25} \right)^{-1} = 14.191 \text{ d}, \quad T^- = \left(\frac{1}{14.765} - \frac{1}{365.25} \right)^{-1} = 15.387 \text{ d}. \quad (4)$$

These are fixed entirely by astronomical constants — but they are not a second, independent prediction. Table 1 shows T^+ and T^- reproducing the aliases of O_1 and MK_3 to within 0.0004 d, and identity (3) shows why: substituting $f_c = |a(M_2)|$ and $f_m = |a(K_1)|$ turns (4) into (3) term by term. The residual is rounding and nothing else, arising almost entirely from writing the carrier as 14.765 d rather than the exact alias period 14.765 294 d; the tropical-year-versus- K_1 difference contributes some four percent of it, in the opposite direction. With unrounded constants, (4) returns the O_1 alias to machine precision and the MK_3 alias to 2×10^{-6} d, the latter reflecting only the rounding of the tabulated MK_3 speed. The two routes do not merely agree closely: they are one expression evaluated twice.

The triplet is testable only if the record separates it. Frequency resolution is $\Delta f = 1/n$, or $\Delta T \approx T^2/n$ in the period domain; on the $n = 5,479$ AR(1) innovations used for testing this gives $\Delta T \approx 0.040$ d at 14.765 d, against predicted carrier-sideband separations of 0.574 d and 0.622 d.

5 Methodology: The Triangulation Protocol

Four stages, each answering a question the previous cannot, none treated as decisive alone.

Stage 1: stationarity diagnostics. *Stationarity* covers two questions: whether a stochastic trend is present at the zero frequency, and whether the seasonal pattern is itself stochastic; only the second determines whether the annual cycle may be removed by decomposition. We run two pairs with opposed nulls — ADF and KPSS at the zero frequency, OCSB and Canova–Hansen at the seasonal one, agreement within a pair being stronger evidence than either alone — and compute the seasonal strength $F_S = \max(0, 1 - \text{Var}(R_t)/\text{Var}(S_t + R_t))$ [22]. Results are in Appendix A.

Stage 2: deseasonalisation. We use classical moving-average decomposition at $p = 365$: deliberate rather than optimal, since it is fully determined by p and introduces no smoothing hyperparameters, so it cannot be tuned toward a particular residual spectrum. Endpoint loss of $(p - 1)/2$ each side leaves $5,844 - 364 = 5,480$ residuals. Stage 1 licenses this over differencing: with no unit root at either the zero or the annual frequency, differencing would remove no stochastic trend while injecting a spurious moving-average structure into exactly the band we target. Because the additive-versus-multiplicative choice is itself an assumption, and a multiplicative model divides out the amplitude variation of interest, we run *both* and require every claim to hold under each — the principal robustness control of the study.

Stage 3: spectral estimation with significance control. Three problems precede calling a peak a finding. *Leakage*: the implicit rectangular window of a finite record spreads energy through its sidelobes, so we apply a Hann taper [8]. *Red noise*: an AR(1) spectrum rises toward low frequencies, so we fit AR(1) and analyse its innovations [17]. *Multiplicity*: we permute the windowed innovations and take upper quantiles of the *maximum* periodogram value over 5 d to 100 d across $B = 1000$ permutations as family-wise thresholds, needing no Bonferroni correction; a minimum separation constraint prevents adjacent bins of one broadened peak being reported twice.

Stage 4: parametric physical validation. The final stage compares the data against the parameter-free predictions of Section 4 rather than extracting more from the record. That the two predictions coincide is itself a result, treated as one in Section 6.3.

6 Results

6.1 The raw series conceals the structure

None of the three standard diagnostics applied to the raw series recovers fortnightly structure, and each fails differently. The ACF fails because the annual component contributes autocorrelation at every lag, swamping smaller superimposed oscillations; the PACF because salinity is strongly persistent, so lag 1 absorbs the bulk of the predictable variation and later values fluctuate near zero; the unwindowed periodogram because the annual peak exceeds everything

Table 2. Family-wise permutation test on AR(1)-pre-whitened residuals, band 5 d to 100 d, $B = 1000$. Magnitudes are not comparable across panels; the ratio to t_{99} is scale-free. Significant periods are quoted to three decimals, the precision at which they are Fourier bin centres.

Period (d)	Magnitude	Ratio to t_{99}	Outcome
<i>Multiplicative residual</i> ($t_{95} = 10.22$, $t_{99} = 11.10$)			
14.768	28.20	2.54	$p < 0.01$
15.390	15.44	1.39	$p < 0.01$
14.194	12.98	1.17	$p < 0.01$
7.42	8.70	0.78	n.s.
7.59	8.49	0.76	n.s.
<i>Additive residual</i> ($t_{95} = 103.23$, $t_{99} = 110.95$)			
14.768	280.15	2.52	$p < 0.01$
14.194	125.28	1.13	$p < 0.01$
15.390	116.80	1.05	$p < 0.01$
5.64	93.47	0.84	n.s.
7.42	84.79	0.76	n.s.

else by roughly an order of magnitude, compressing sub-annual components against the axis. This is a finding rather than a preamble: it explains why the reading of [2] was ambiguous — the expected outcome of these diagnostics on an undecomposed estuarine series, not a property of the data — and establishes that decomposition is a prerequisite rather than a refinement, the structure below being not merely clearer after deseasonalisation but inaccessible before it. Deseasonalisation does not remove day-to-day persistence, which is why Stage 3 pre-whitens.

6.2 Significance testing

Table 2 reports the five strongest peaks in the 5 d to 100 d band under each decomposition against family-wise thresholds from $B = 1000$ permutations, and Figure 1 resolves the fortnightly band against those thresholds.

The same three peaks clear the 1% threshold under both decompositions, and are the *only* peaks anywhere in the band to clear even the 95th percentile: the triplet is not the most prominent part of the residual spectrum here, it is the whole of its significant content. The strongest robustness evidence is the ratio column. Raw magnitudes differ by an order of magnitude between panels, a multiplicative residual being dimensionless and centred near unity while an additive one carries the units of the series; normalised by their own thresholds, the dominant peak stands at $2.54\times$ under one model and $2.52\times$ under the other. Two decompositions with different assumptions, on different scales, computed independently, agree to within 0.02 on a scale-free measure of prominence — a test the result could readily have failed.

Proposition 1 additionally predicts *equal* sidebands. The flanking peaks are unequal, by 19% and 7% under the two models, with the direction reversing

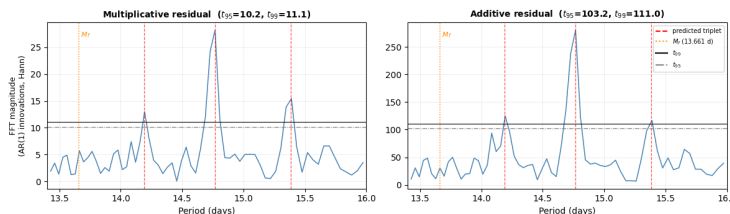


Fig. 1. The fortnightly band resolved. AR(1)-pre-whitened, Hann-tapered periodogram over 13.4 d to 16.0 d, multiplicative (left) and additive (right). Dashed lines mark the predicted positions, the dotted line M_f at 13.661 d, horizontal lines the family-wise 95th and 99th percentiles. All three predicted positions carry peaks clearing the 99th percentile; M_f does not.

Table 3. Observed triplet positions against the two parameter-free predictions: the modulation column applies Proposition 1 to M_{sf} and the tropical year, the alias column reads off Table 1. $|\Delta|$ is against the modulation prediction, ΔT the bin width.

Component	Modulation (d)	Alias (d)	Observed (d)	$ \Delta $ (d)	ΔT (d)
Upper sideband	14.191	14.192 (O_1)	14.194	0.003	0.037
Carrier	14.765	14.765 (M_2)	14.768	0.003	0.040
Lower sideband	15.387	15.387 (MK_3)	15.390	0.003	0.043

between them; at this precision the asymmetry neither supports nor contradicts modulation, and under the aliasing account no symmetry is expected at all, since O_1 and MK_3 carry independent amplitudes.

One prediction does discriminate, and it fails to materialise. Because the daily maximum rectifies the tide, difference frequencies should appear directly; the leading diurnal difference $f_{K_1} - f_{O_1}$ corresponds to 13.661 d — the lunar fortnightly period M_f , and the frequency at which the diurnal inequality, hence which daily high water is the larger, cycles. A triplet manufactured by the aggregation convention should deposit energy there as readily as at 14.765 d. It does not: the largest magnitude within one bin of 13.661 d reaches only $0.51 \times$ the 99th percentile under the multiplicative model and $0.28 \times$ under the additive one, with no local peak at all under the latter. Whatever populates the triplet, it is not a generic nonlinear artefact of taking daily maxima. The remaining sub-significant peaks lie near 7.4 d in both panels, together with one at 5.64 d under the additive model. The 7.4 d candidates sit close to half the fortnightly values, but their offsets exceed the resolution there ($\Delta T \approx 0.010$ d) by several bins, and we decline to interpret them.

6.3 Parametric validation, and the limit of what it can settle

All three observed peaks fall within one bin of prediction, agreeing to 0.003 d — about a tenth of the bin width — though a periodogram cannot localise a

peak more finely than one bin, so three-decimal agreement is not three-decimal accuracy. What Table 3 establishes is that a prediction made without reference to the data landed in the correct bin three times out of three.

That prediction does not identify a mechanism, and we say so rather than let the quality of the fit imply otherwise. The aliases of O_1 and MK_3 fall at 14.1916 d and 15.3873 d, and identity (3) shows this is not proximity but identity, the two predictions differing only through rounding of the carrier period. “A carrier with sidebands separated by the annual frequency” and “the aliases of O_1 , M_2 and MK_3 ” are the same arithmetic statement, so the closeness of the agreement is not evidence for one account over the other but a consequence of their being numerically identical.

We therefore state the finding at the level the data support. The triplet is present, robust across decomposition models, and separated in frequency by exactly the annual cycle; whether that separation arises from discharge-driven annual modulation, from the aliased superposition of three constituents, or from both, is not decidable at daily resolution. Nor does the daily-maximum convention break the tie, although it might appear to: a maximum does not alias in the textbook sense, but it depends on each constituent’s phase at the instant of the daily peak, and that phase advances by precisely the alias frequency of (2). The same three periods are inherited either way. Resolving the question requires sub-daily observations, not a longer daily record.

7 Discussion

Evaluation design. Under either reading the fortnightly signal varies in strength across the year, so a single aggregate error metric averages a strong regime against a weak one and obscures performance in the dry-season window that matters operationally. Seasonal stratification is warranted; aggregate reporting is the prevailing practice in the studies of Section 2.

Transferability. The quantities generating the predictions of Section 4 are astronomical constants, independent of station, estuary and continent, so the predicted positions hold at any site with a mixed semi-diurnal regime and an annually varying discharge; only amplitudes, hence detectability, differ. They are thus prior knowledge rather than quantities to be searched for — and the identification ambiguity is equally general, since any study sampling an estuarine variable daily inherits it. The amplitudes carry their own prediction: moving landward, bottom friction damps the tide while discharge takes an increasing share of the salt balance, so the flanking-to-carrier ratio should decline along a transect — a candidate spatial diagnostic measurable with this protocol as it stands. All results derive from one station, so portability is argued from the physics rather than demonstrated.

7.1 Why spectrum-preserving surrogates cannot test for amplitude modulation

Amplitude modulation is commonly assessed by band-pass filtering around the carrier, extracting the analytic envelope via the Hilbert transform, and testing the envelope spectrum at the modulating frequency, with the phase-randomised surrogate [21] as the apparently natural null. We applied it and obtained non-significant outcomes under both decompositions: the observed statistic reaches $0.83\times$ the 99th percentile under the multiplicative model (49.92 against 60.43, at 365.3 d) and $0.50\times$ under the additive one (249.39 against 494.46, at 322.4 d).

The result is not evidence against modulation, because the test cannot produce evidence either way. By Proposition 1 an amplitude-modulated carrier is *completely* characterised, in the linear power spectrum, by three components at f_c and $f_c \pm f_m$; phase randomisation preserves $|X(f)|$ at every frequency, so every surrogate retains all three at their observed magnitudes. With arbitrary randomised phases $\varphi_-, \varphi_0, \varphi_+$, the band-passed surrogate $a_0 \cos(2\pi f_c t + \varphi_0) + a_- \cos(2\pi(f_c - f_m)t + \varphi_-) + a_+ \cos(2\pi(f_c + f_m)t + \varphi_+)$ has an analytic envelope containing beat terms at f_m and $2f_m$ for *any* phases. Envelope modulation at the modulating frequency is thus present in every surrogate at magnitude comparable to the observation: the null contains the alternative, and the test has no power against it by construction. This also explains why the observed statistic falls *below* the null quantiles — anomalous under a correctly specified test, expected here, since randomised phases can produce larger excursions than the configuration realised in the data.

The point generalises, and compounds the identification problem of Section 6.3. Spectrum-preserving surrogates are the appropriate null for *nonlinear* structure, which is what they were introduced for [21]; a modulated carrier and an aliased constituent triplet have identical second-order spectra by construction, so no test built on that spectrum — surrogate or parametric — can separate them.

8 Feature Specification

The consequence for forecasting survives the identification ambiguity, depending on the observed structure rather than its cause. The relevant structure is a triplet, so a feature set encoding one fortnightly component — a single lag at fifteen days, or one pair of Fourier terms at the fortnightly frequency — represents a third of it. With t indexing days and $\text{doy}(t)$ the day of year, the *triplet basis* adds to whatever lag features a model already uses the six terms

$$\left\{ \cos(2\pi f_j t), \sin(2\pi f_j t) \right\}_{j \in \{-, 0, +\}}, \quad f_- = \frac{1}{15.387}, \quad f_0 = \frac{1}{14.765}, \quad f_+ = \frac{1}{14.191}, \quad (5)$$

the three frequencies fixed *a priori* by Table 3 and estimated from nothing. The natural ablations are a *carrier-only* basis retaining the $j = 0$ pair, and a

carrier-plus-phase basis retaining that pair together with $\cos(2\pi \text{doy}(t)/365.25)$ and $\sin(2\pi \text{doy}(t)/365.25)$, which encode position in the annual cycle without encoding the sidebands.

The validation design repeats no part of this analysis: train a standard sequence learner — an LSTM of the kind used by [12], or TimesNet [23] — under each basis on an unchanged lookback and horizon grid, with a chronological split and error reported separately for the dry and wet seasons. No gain from the triplet basis would indicate that the flanking components carry no predictive information despite being spectrally present; a gain matched by carrier-plus-phase, that what they carry is seasonal position, obtainable more cheaply; a gain exceeding it, structure not reducible to a seasonal index.

A second consequence answers the question the paper opened with. A fortnightly lookback spans one cycle of the dominant component, which is why grid search keeps selecting it, but cannot resolve structure separated from it by the annual frequency: a modulation of period 365 d is not recoverable from within a 15 d window containing no complete cycle of it. The information is not merely hard to extract, it is absent, and supplying it exogenously costs two features. These are specifications rather than results: (5) is stated so that the prediction is falsifiable by any reader with a forecasting pipeline.

9 Limitations and Threats to Validity

Daily sampling prevents attribution. The aliases of O_1 , M_2 and MK_3 fall at the three observed periods and are indistinguishable from a fortnightly carrier with annual sidebands. Our results establish that all three peaks are of tidal origin and separated by exactly the annual frequency; they do not establish which route deposits the energy, and discriminating requires sub-daily sampling this record does not provide.

Sideband significance is marginal under a white null. The permutation procedure destroys all temporal structure, so it tests against a white null that AR(1) pre-whitening only partly licenses: Ljung–Box rejects whiteness of the AR(1) innovations at every lag we checked ($p < 10^{-27}$). Under a circular block bootstrap — which preserves short-range dependence but, at block lengths short compared with a fortnight, cannot reproduce a fourteen-day cycle — the carrier remains unambiguous at $2.1\text{--}2.3\times$ the 99th percentile, while the flanking peaks fall to $0.97\text{--}1.23\times$, significant but slender. Under the AIC-selected AR(18) filter only the carrier survives; we do not read this as refutation, since an eighteen-lag filter can represent the oscillation it is meant to whiten, and its order was in part selected because of that structure. Readers who prefer this more aggressive null should treat the sidebands as unproven.

Impact of Imputation. Although gap-filling was applied to 1.64% of the analysed window, this minimal fraction is insufficient to manufacture the robust, high-amplitude fortnightly triplet observed in the spectrum. Nevertheless, minor

artefactual damping of amplitudes or the generation of spurious sub-significant peaks (such as those near 7.4 d) cannot be strictly ruled out.

No downstream validation, and further caveats. No forecasting model was trained and no predictive metric computed, so Sections 7 and 8 state hypotheses rather than demonstrated gains. Conditional heteroskedasticity was not tested and the residual shows elevated dispersion around the most extreme dry seasons in the record, so the record is not homogeneous across its span, though this does not invalidate a maximum-statistic permutation test. Classical moving-average decomposition also cannot accommodate a seasonal pattern whose shape evolves, a constraint a LOESS-based alternative would relax at the cost of smoothing hyperparameters this design avoids.

10 Conclusion and Future Work

A hyperparameter the forecasting literature has been selecting by search turns out to have a physical referent, and examining it reveals a richer structure than the search could find. A grid over window lengths can establish that roughly a fortnight works; it cannot report that the fortnight is the observable image of the tide, nor that two further resolvable components accompany it at exactly the annual offset. The second lesson concerns what comparison against theory buys. Two astronomical derivations — annual modulation of the spring-neap beat, and aliasing of O_1 , M_2 and MK_3 — predict the same three periods, because the tidal frequency identities survive aliasing while K_1 aliases onto the year. An extremely close fit to a parameter-free prediction is therefore compatible with either mechanism, and a study reporting only the fit would have overstated what it had shown. External prediction constrains positions rather than mechanisms: where two mechanisms share a second-order spectrum, no spectral test will separate them.

Three directions follow. Resolution: repeating the analysis on sub-daily records, which would separate O_1 , M_2 and MK_3 and settle the attribution. Downstream validation of Section 8: training standard sequence architectures on the triplet basis against the carrier-only and carrier-plus-phase ablations, with seasonally stratified evaluation. And spatial extension: applying the protocol across a transect to see how the flanking amplitudes decay from the mouth under bottom friction and increasing freshwater dominance, which would turn the flanking-to-carrier ratio into an interpretable spatial feature and, incidentally, discriminate between the mechanisms, since only one predicts that the ratio tracks the discharge gradient.

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A Stationarity Diagnostics and Reproducibility

All five Stage 1 tests agree. ADF rejects (-6.9753 , $p = 8.46 \times 10^{-10}$); KPSS fails to reject under both `c` and `ct` (0.0410, 0.0373; $p > 0.10$, inequalities because the lookup range is exhausted); OCSB and Canova–Hansen both return $D = 0$. Seasonal strength is $F_S = 0.882$ (additive) and 0.863 (log scale), against the 0.64 reference [22] — very strong yet deterministic, which licenses decomposition over differencing.

Series: daily maximum salinity, Vam Kenh, 2009-01-01 to 2024-12-31. **Stationarity:** ADF with AIC lag selection; KPSS with automatic lag selection under `c` and `ct`; seasonal period $m = 365$. **Decomposition:** classical moving-average at period 365, additive and multiplicative, leaving 5,480 residuals. **Spectral estimation:** AR(1) pre-whitening leaving 5,479 innovations; Hann taper; band 5 d to 100 d; peak separation 5 bins; $B = 1000$ permutations; seed 42; observed peak positions are Fourier bin centres n/k with $k = 386, 371, 356$, and $\Delta T \approx T^2/n$. **Targeted queries:** for each period T_0 of Table 1 and for $M_f = 13.661$ d, the reported magnitude is the maximum of the innovations periodogram over $|T - T_0| \leq T_0^2/n$, with a peak recorded only if `scipy.signal.find_peaks` (separation 5 bins) returns one there. **Envelope analysis:** fourth-order Butterworth band-pass over 13.5 d to 16.0 d, zero-phase; Hilbert envelope; modulation band 300 d to 400 d; 500 phase-randomised surrogates. **Constituent speeds** (degrees per mean solar hour): $M_2 = 28.9841042$, $S_2 = 30.0000000$, $K_1 = 15.0410686$, $O_1 = 13.9430356$, $MK_3 = 44.0251729$; tropical year 365.25 d. **Software:** Python 3.13.12, numpy 2.1.3, pandas 2.3.2, scipy 1.16.3, statsmodels 0.14.5 [20], pmdarima 2.1.1, matplotlib 3.10.6, openpyxl 3.1.5.

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